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Enhancing Medical Visual Question Answering with Explainable Knowledge Graphs and Agentic RAG

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***Abstract*—Medical Visual Question Answering (VQA) systems are crucial for supporting clinicians in interpreting medical images; however, their lack of transparency hinders their adoption in clinical practice. This paper presents a novel multi-modal framework that integrates Large Language Models (LLMs) with Retrieval-Augmented Generation (RAG) and Knowledge Graphs (KGs) to improve the clarity and reliability of medical VQA systems. By leveraging the R-RAD dataset, our approach generates rationale-backed responses, ensuring that the reasoning process is explicit and trustworthy for medical professionals. The proposed method achieves state-of-the-art performance on the RRAD benchmark, with an accuracy of 85.57% on closed-ended questions and robust rationale generation, as evidenced by a BERTScore F1 of 87.82%. This work demonstrates that medical VQA systems can be both highly accurate and interpretable, facilitating their practical use in healthcare settings.**

***Index Terms*—Medical VQA, Knowledge Graphs, RAG, Agentic RAG**

I. Introduction

Medical Visual Question Answering (MedVQA) systems, designed to diagnose and answer questions about medical images, face a vital challenge: their outcomes often lack transparency, which damages the trust demanded for clinical practice. Nearly all of the public MedVQA datasets do not provide explanations of how and why these questions were answered like this, making it difficult to assess a model's performance. This results in a low-transparency system that physicians would hesitate to use in real healthcare environments. While more recent datasets contain medical images, questions, and answers, they lack detailed diagnostic reasoning. Our research introduces a method to enhance the clarity and reliability of medical VQA systems. The contribution of this study is as follows:

- Hybrid KG-RAG Retrieval, integrating knowledge graphs with RAG to enhance performance on the R-RAD dataset by enhancing response precision and rationale accuracy.
- Generates rationale-backed responses using the Agentic RAG system, causing generated responses to be more reasonable, particularly for open-ended questions.
- Achieves the highest score on the R-RAD benchmark, setting a new standard for medical VQA.

We aim to demonstrate that medical VQA systems can be precise and straightforward to comprehend, thereby enhancing transparency and confidence in medical decision-making.

II. Related Work

Recent efforts have concentrated on creating large-scale medical QA datasets to advance AI in healthcare. The efforts included datasets and benchmarks that incorporate both vision and text in clinical QA systems.Moreover, some advanced work covered enhanced VQA with Rationales for further reasoning and explainability in VQA tasks.

Large-scale textual resources like Huatuo-26M [1], with its 26 million QA pairs, have been pivotal for text-based medical AI. However, its text-only nature makes it inapplicable to multimodal VQA, underscoring the need for image-integrated datasets. In contrast, PathVQA [2] benchmark advanced multimodal research with 30,000+ QA pairs from pathology images. Its specialization in pathology, however, contrasts with the broader radiological scope required by our work.

Recent efforts have prioritized interpretability and diagnostic integration in MedVQA for basic clinical needs. The work by [3] introduced multimodal rationales—combining visual saliency maps with textual explanations—to justify model answers, However, its dependence on annotated rationales limits scalability to some open-ended questions. Building on this, [3] proposed an enhanced framework for medical VQA with multimodal determination rationales, However, these approaches' dependence on annotated rationales limits scalability to some open-ended questions. Meanwhile, [4] leveraged LLM-generated explanations to enable zero-shot diagnosis, bridging the gap between image analysis and textual reasoning without task-specific training. While RAG and Knowledge Graphs (KGs) are used in medical AI, it is unclear how to best combine them to generate reliable explanations for VQA. Furthermore, the effectiveness of newer "agentic" systems has not been properly tested for high-stakes medical tasks. Our work addresses these two key gaps. Future work should focus on document-enhanced VQA systems that integrate clinical context with multimodal data, coupled with better generalization, as table I compromises. This shift could bridge the gap between research benchmarks and real-world clinical decision support in VQA.

III. Methodology

This section outlines a thorough methodology employed to generate trustworthy responses along with their rationale using the R-RAD dataset. The proposed pipeline, illustrated in Fig-

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TABLE I: Comprehensive Medical VQA Literature Review

Work	Contribution	Modality	Size	Limitations	Clinical Use
Huatuo-26M [1]	Largest Chinese medical QA	Text-only	26M QA pairs	No visual data	Broad symptom coverage
PATHVQA [2]	Pathology VQA benchmark	Image+Text	30K QA pairs	Narrow specialty focus	Cancer diagnosis aid
Multimodal Rationales [3]	Explainable predictions	Image+Text	5K annotated cases	Labor-intensive	Medical education
Enhancing Medical VQA [3]	Multimodal determination rationales	Image+Text	8K cases	Complex annotation needs	Diagnostic justification
ChatGPT Framework [4]	Zero-shot diagnosis	Image+Text	N/A	Hallucination risks	Preliminary screening

ure 1, integrates advanced techniques from knowledge graph construction, embedding and retrieval, hybrid approaches, and agentic RAG to assure contextually relevant and precise responses. The methodology leverages the processed R-RAD dataset and employs Neo4j for the knowledge graph, FAISS for vector storage, and LLMs for answer generation. The system implementation follows four key stages. First, a structured Knowledge Graph is built from the R-RAD dataset using Neo4j, capturing relationships between images, questions, answers, and medical concepts. Second, CLIP embeddings for images and text are computed and stored in FAISS, enhancing retrieval via similarity search. Third, a hybrid retrieval approach integrates both vector-based (RAG) and KG-based methods to fetch and rank relevant context for each query. Finally, an agent-based reasoning module retrieves relevant data and generates accurate, multi-step responses using a large language model. This pipeline leverages the strengths of KGs for structured data representation, embeddings for semantic retrieval, and agentic RAG for dynamic answer generation, culminating in a robust framework for medical question-answering tasks.

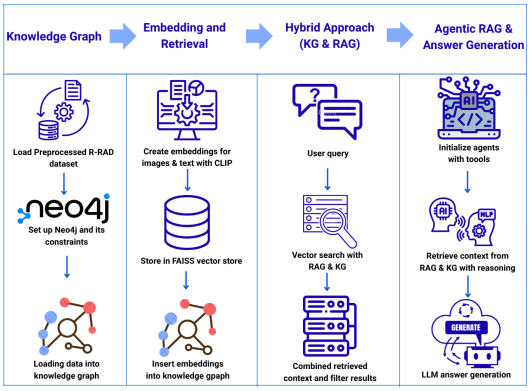

Fig. 1: Proposed Pipeline

A. *Dataset Description*

In our implementation, we worked with the R-RAD [3] dataset. The availability of thorough, human-written rationales alongside each image and QA pair makes it an ideal benchmark for evaluating our system. The R-RAD dataset is divided into training and testing sets, which include both open-ended and closed-ended questions. The training set includes 3,064 samples, with 1,823 closed-ended and 1,241 open-ended questions. The test set contains 451 samples, with 272 closed-ended and 179 open-ended questions. Each sample incorporates a question, answer, an associated medical image, the organ depicted (e.g., chest, head, abdomen), and a detailed solution (rationale) explaining the answer. The images consist primarily of chest X-rays, MRI scans, or CT scans. Figure 2 shows a sample of the R-RAD dataset.

Question: Are the lungs normal appearing?
Answer: no
Solution: The provided chest x-ray shows some abnormalities indicating that the lungs are not normal appearing. There is evidence of a density seen on the left side of the image, which corresponds to the right lung of the patient due to the orientation of the x-ray. This density suggests the presence of some pathological process, such as an infection, mass, or consolidation, which is not normal in a healthy lung. The normal appearance of the lung on an x-ray should show clear lung fields with well-defined vascular markings, and no unusual opacities or masses should be present.

Fig. 2: Sample of R-RAD Dataset

B. *Knowledge Graph Construction*

The Knowledge Graph (KG) component of our model pipeline (Figure 1) is built using Neo4j to manage and represent structured knowledge from 3,064 radiology QA records. A summary of the ingestion workflow is provided in Figure 3.

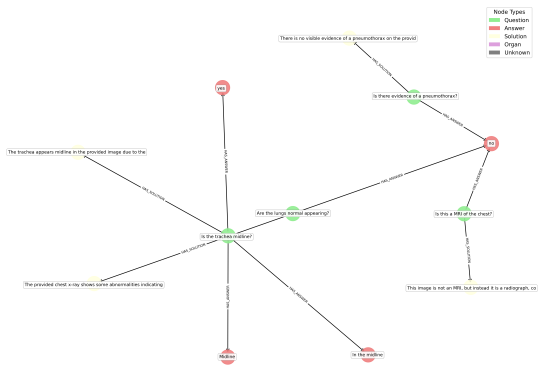

Fig. 3: The Knowledge Graph combining images and texts

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C. *Embedding & Retrieval*

The Embedding & Retrieval stage, as shown in Figure 1, employs the CLIP model [5] to generate embeddings for images and text from the R-RAD dataset. Images are pre-processed by resizing to 512x512 pixels and converting to RGB, while questions are tokenized for text embedding. Each image and question is embedded individually, producing 768-dimensional vectors that capture semantic content. These embeddings are stored in a FAISS vector database, allowing efficient similarity searches. The embeddings are also integrated into the knowledge graph as node properties, allowing the system to combine structured and semantic retrieval.

D. *RAG*

The Hybrid Approach - KG & RAG stage processes user queries by leveraging both RAG and KG retrieval mechanisms. Upon receiving a query, the system performs a vector search using the FAISS index to identify semantically similar images and questions based on their CLIP embeddings. Simultaneously, the knowledge graph retrieves structured context, such as organ details and solutions, using Cypher queries. The retrieved contexts from RAG and KG are combined, ranked based on relevance, and filtered to eliminate noise, ensuring a comprehensive and accurate context set. This integration capitalizes on RAG's flexibility in semantic matching and KG's precision in structured data, addressing diverse query types (e.g., exact organ queries vs. vague condition descriptions) despite using the same underlying data.

E. *Agentic RAG*

Moving to the agentic RAG system phase, as shown in the pipeline 1, implemented using LangChain [6], where the Google Gemini [7] model serves as the central reasoning agent. Tools, which are defined by LangChain's @tool decorator, enable the agent to dynamically interact with the knowledge base for the retrieval of information and the generation of answers.

In our implementation, we designed four retrieval tools to support case analysis and knowledge access. Starting with the first tool, *Textual Case Similarity Search tool* uses CLIP [8] text embeddings and FAISS indexing to retrieve semantically similar medical cases with detailed metadata. *Visual-Semantic Image Retrieval tool* matches text queries to medical images based on visual semantics, returning ranked results. *Knowledge Graph Interrogation tool* enables structured retrieval from the Neo4j graph via Cypher queries, keeping both image and text-based searching. Finally, the *General Graph Query Interface tool* allows for adaptable, custom Cypher queries, allowing advanced exploration of the graph.

Then, the agent execution framework is as follows: The agent is guided by the ReAct (Reasoning and Acting) framework [9], a prompting technique created to produce step-by-step reasoning and tool use from LLMs. For our implementation, we employ a standardized prompt template based on this framework that guides the agent to think sequentially (Thought, Action, Action Input, Observation). The prompt is also customized via this prompt template shown in Figure 4 to include specific instructions about the VQA task, knowledge base schema, radiological conventions, and the required output format for the final answer.

Prompt

Follow the ReAct format: include `Thought:`, `Action:`, and `Action Input:` for each step. Do not skip `Action:` after `Thought:`. If no action is needed, write `Action: None`.

For closed-ended questions (`answer_type=CLOSED`), answer `yes` or `no`. For open-ended questions (`answer_type=OPEN`), provide a descriptive response.

Analyze the provided medical image to identify features & combine this with context & knowledge graph data.

Use `query_knowledge_graph` with Cypher queries targeting `Question` or `Image` nodes (e.g., `MATCH (q:Question) WHERE q.text CONTAINS consolidation`).

The rationale must explain the reasoning, referencing image analysis, context, and knowledge graph data.

Fig. 4: Custom Prompt Including Specific Instructions About The VQA Task

The agent is built using Langchain's create_react_agent, combining a model, tool suite, and ReAct prompting. Upon receiving a query, image analysis from Gemini, and answer type, the agent enters a ReAct reasoning loop: it thinks, selects tools, observes results, and iterates until sufficient context is gathered.

F. *Answer Generation*

The answer generation process in this system is a refined multi-stage pipeline. Its goal is to generate an accurate and reliable response to medical Visual Question Answering (VQA) questions by merging initial visual understanding with reasoned knowledge retrieval and reasoning. This process involved using two different LLMs.

a) Gemini LLM

Gemini [7] is a multimodal model family by Google DeepMind, with gemini-1.5-flash optimized for fast, efficient reasoning across text, images, code, and audio. In the VQA pipeline, Gemini is used in two parts: first, to generate brief visual summaries of medical images based on user questions, and second, as the core reasoning engine within the Langchain agent for downstream processing.

b) Deepseek LLM

DeepSeek models [10] are known for strong instruction-following, coding, and logical reasoning. In this pipeline, the deepseek-r1 model functions as the core synthesis engine, generating final answers based on the user's query and hybrid contexts retrieved via RAG+KG. It is prompted to act as an expert medical VQA assistant, reasoning strictly from the provided information.

IV. Results and Discussion

This section will discuss the evaluation metrics used in our implementation and investigate the performance of the two different experiments, LLM with KG & RAG and LLM with KG and Agentic RAG.

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A. Evaluation Metrics

To evaluate the performance of our generated responses, open-ended & closed-ended, we evaluated the closed-ended questions using the Accuracy metric to measure the accuracy of predicted responses and BERTScore-F1 [11] to evaluate the semantic similarity of generated rationales compared to the ground truth rationales. Whereas for the open-ended questions, both the rationale and the generated responses were evaluated using BERTScore [11], BLEU [12], and Rouge [13] metrics.

B. Experimental Results

This section details the experimental findings on closed-ended and open-ended questions.

a) RAG System Performance

KG + RAG with DeepSeek achieves the highest accuracy on closed-ended questions (85.57%), outperforming both MedThink (83.5%) and Gemini-based models. It also produces the most accurate rationale explanations with a BERTScore F1 of 87.82%.

For open-ended questions, KG + RAG (DeepSeek) leads in ROUGE-1 recall (57.59%) and ROUGE-L recall (33.38%), reflecting strong coverage and coherence. While MedThink scores higher in ROUGE-2 (20.2%) and BLEU-4 (8.8%), DeepSeek remains competitive with a BLEU-4 of 3.13%.

In contrast, the Gemini-based KG + RAG setup shows lower performance, because Gemini depends more on its training knowledge context than the retrieved context, causing it to achieve a lower accuracy, with a 76.84% accuracy and 83.53% BERTScore on closed-ended tasks. It also underperforms on open-ended questions, with ROUGE-1 recall at 45.75% and BLEU-4 at only 0.99%.

b) Agentic RAG System Performance

The Agentic RAG version using Gemini performs the worst overall. This is because the task is considered simple and does not require multi-step agentic reasoning, as adding agentic sophistication can increase errors and lower the precision and the reasoning exactness. On closed-ended questions, it records 62.59% accuracy and a rationale BERTScore F1 of 58.72%, suggesting limited benefit from multi-step reasoning.

For open-ended tasks, it achieves a ROUGE-1 recall of 23.78% and BLEU-4 of 2.42%. Although the BLEU-4 slightly exceeds the Gemini KG + RAG setup, its overall rationale quality remains poor.

V. Conclusion

This work advances medical VQA by combining knowledge graphs, semantic embeddings, hybrid retrieval, and agentic RAG, using the R-RAD dataset. The KG + RAG system with DeepSeek achieved 85.57% accuracy and strong rationale quality (BERTScore F1: 87.82%). Conversely, our results challenge the assumption that complex agentic reasoning is always superior; the Agentic RAG underperformed significantly (62.59%), indicating that simpler, direct retrieval is more robust for this clinical task. Key limitations like dataset size and computational cost remain, and future efforts should focus on scaling data and refining agent design to support more clinically viable VQA systems.

TABLE II: Performance on Closed-Ended Questions

System	Accuracy	Rationale BERTScore F1
MedThink [3]	83.5%	–
KG + RAG (DeepSeek)	**85.57%**	**87.82**
KG + RAG (Gemini)	76.84%	83.53
Agentic RAG (Gemini)	62.59%	58.72

TABLE III: Rationale Performance on Open-Ended Questions

System	ROUGE-1 Recall	ROUGE-2 Recall	ROUGE-L Recall	BLEU-4
MedThink [3]	**50.2**	**20.2**	**29.5**	**8.8**
KG + RAG (DeepSeek)	**57.59**	17.21	**33.38**	3.13
KG + RAG (Gemini)	45.75	14.34	44.39	0.99
Agentic RAG (Gemini)	23.78	7.12	15.33	2.42

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